

NICOLAS EDMOND

ML / AI Engineer

📍 Bordeaux, France

☎ +33 7 83 26 53 18

✉ nicolasedmond8@gmail.com

🌐 LinkedIn

🐙 GitHub



ABOUT

AI Engineer focused on architecting robust, production-grade agentic systems. I specialize in designing multi-agent workflows, hybrid RAG pipelines, and high-performance backends. I am committed to staying at the forefront of AI research, continuously exploring emerging models and optimization techniques to solve complex real-world problems. Driven by a methodology forged in endurance sports, I prioritize architectural purity, system reliability, and minimizing latency in LLM-powered infrastructures.

EXPERIENCE

Co-CTO - AI & Agentic Engineer | Formalis.IA

Since Sep 2025

Bordeaux



- **Agentic Workflow & LegalTech Automation:** Architected a deterministic **multi-agent workflow** (**LangGraph** + **GCP Vertex AI**) with 4 sequential nodes (Classifier, Detector, Mapper, Validator) to autonomously classify unstructured legal PDFs, detect **11 distinct corporate modification types**, and map files.
- **High-Performance Extraction:** Engineered parallel **S3 document processing** pipelines (**ThreadPoolExecutor**) with advanced prompt engineering to handle edge cases and transform raw uploads into strict JSON payloads. This completely replaced legacy systems and drastically reduced manual review time.
- **Hybrid Cloud Orchestration & Persistence:** Built a scalable event-driven orchestrator on **AWS Lambda** (Docker) managing fault-tolerant operations (AWS ↔ GCP). Integrated **DynamoDB** state tracking for real-time observability, and automated **INPI/RNE** API submissions using **AWS SAM**.
- **Monitoring & Automated Validation:** Implemented a **DynamoDB-backed tracker** for real-time observability and a robust automated testing suite (pytest), ensuring high availability for critical legal document workflows.
- *Stack: Python 3.12, LangGraph, AWS (Lambda, DynamoDB, S3, SAM), GCP Vertex AI, Docker, Prompt Engineering.*

Data Engineer | RTE

Sep 2022 – Sep 2025

Lille



- Engineered **FIDEL (Shiny/R)**, a predictive analytics platform leveraging **Time Series Forecasting** (ARIMA/SARIMA) and **Machine Learning** to deliver localized wind predictions for **300+ wind farms**, enhancing grid balancing accuracy by **15%**.
- Architected and deployed large-scale **ETL pipelines** (PostgreSQL, Oracle) processing **thousands of daily telemetry data points**, eliminating **200+ hours** of manual data wrangling annually and accelerating predictive modeling workflows.
- *Stack: Python, R, Shiny, PostgreSQL, Oracle, Data Lake, ARIMA/SARIMA, Machine Learning.*

PROJECTS

Agentic Risk Assessment Framework

- Architected a production-grade multi-agent pipeline (**LangGraph**) using **Domain-Driven Design (DDD)**, orchestrating 3 autonomous agents to synthesize CRO-level risk reports in **<60 seconds** — seamlessly switching between local models (**Ollama**) and cloud APIs (**Gemini 2.5 Flash**).
- Engineered a **Hybrid RAG** retrieval pipeline combining semantic vector search (**PostgreSQL/pgvector**) and BM25 keyword matching, fused with **Reciprocal Rank Fusion (RRF)** for high-precision corporate disclosure retrieval.
- Developed a deterministic supervisor with LLM self-correction and a **Reinforcement Learning (RL) feedback loop**, dynamically adjusting data source weights in **PostgreSQL** based on user validation.
- Implemented resilient state management & **async streaming**, utilizing **Redis** checkpointing, **asyncio**, and thread-safe queues to broadcast live agent execution logs to a **Streamlit** frontend.
- *Stack: Python, LangGraph, Gemini 2.5, PostgreSQL, pgvector, Redis, Streamlit, Plotly, Asyncio, Docker, Ollama.*

Multimodal Alpha-Signal Extractor (Vision-Language Trading Pipeline)

- Architected an end-to-end multimodal pipeline using **Domain-Driven Design (DDD)**, fine-tuning **Qwen2.5-VL-3B** via **Unsloth (QLoRA)** to analyze synthetic financial candlestick charts and generate structured trading signals.
- Engineered a custom **synthetic dataset generation** pipeline (`yfinance`, `mplfinance`) to automatically pair technical charts (RSI, Bollinger Bands) with context-aware market news and forward-return ground truths.
- Built an **alignment pipeline** for multimodal models, implementing **DPO (Direct Preference Optimization)** via TRL to align the model's visual reasoning with realized market returns, directly increasing prediction confidence calibration.
- Deployed the fine-tuned VLM to production via **Hugging Face Spaces** using **llama.cpp** (GGUF quantization) and orchestrated a dual-agent system (VLM + Sentiment LLM) with **LangChain** and **Streamlit** for real-time inference.
- *Stack: Python, PyTorch, Unsloth, TRL (DPO), Qwen2.5-VL, llama.cpp, LangChain, Streamlit, Pandas, yfinance.*

PCBA Cost Prediction Pipeline | Schneider Electric



- Engineered an end-to-end ML pipeline predicting manufacturing costs, achieving an accurate **MAPE of 9.05%** (down from 15.6%) and **R² of 0.972**.
- Designed a robust **Stacking Ensemble** (XGBoost, RF, GBM, Ridge) coupled with target log1p-transformation to process complex distributions dynamically.
- Improved reliability by eradicating data leakage and establishing a resilient preprocessing workflow (**IQR filtering, RobustScaler, indicator-based imputation**).
- Systematically optimized model stability through CV and hyperparameter tuning. *Stack: Python, scikit-learn, Pandas, XGBoost, Ensemble Learning.*

EDUCATION

Specialized Master in Big Data | GEM & Ensimag — Grenoble



2025 – 2026

ML, Deep Learning, Large-Scale Data Processing, Customer Knowledge & Marketing.

Engineering (Electrical & Energy) | ESME — Lille



2022 – 2025

Electrical Engineering, Computer Vision, Assembly.

BA1 - Bachelor | EPFL — Lausanne



2021 – 2022

Mechanical Engineering.

Intensive Preparatory Classes (CPGE) | Lycée Faidherbe — Lille

2019 – 2021

Physics, Chemistry, Mathematics (PCSI/PC).

High School Diploma (Scientific) | LFIB — Bangkok, Thailand

2016 – 2019

TECHNICAL SKILLS

Languages: Python, C/C++, SQL, Java, R, Bash, TypeScript / JavaScript, LaTeX

ML / AI: PyTorch, TensorFlow, Hugging Face Transformers, Scikit-Learn, OpenCV, Pandas, NumPy

LLM Ecosystem: LangChain, LangGraph, LlamaIndex, LoRA/QLoRA, vLLM, Unsloth, Ollama

Backend: FastAPI, SQLAlchemy, Pydantic, AsyncIO, Redis, WebSockets, REST APIs

DevOps / Cloud: Docker, Kubernetes, AWS (Lambda, S3, ECS, EC2), GCP (Vertex AI), Terraform, Git, CI/CD, GitHub Actions

Databases: PostgreSQL (pgvector, full-text search), Redis, MongoDB, Oracle, Blob Storage (S3)

Architecture: Clean Architecture, DDD, TDD, Repository/Factory/Ports & Adapters

Research: Mathematical modeling, information theory, Markov chains, Bayesian inference

LANGUAGES & CERTIFICATIONS

Stanford Certifications: Machine Learning & Deep Learning Specializations (2025)

Languages: French (Native) | **English** (Fluent, TOEIC 920/990) | **Spanish** (Conversational, DELE B1)

ASSETS & INTERESTS

Soft Skills: Eloquence & Rhetoric (Persuasive comms) | **Team Spirit** | **Determination & Rigor**

Volunteering: Vice Head of Communications at **ESME Sans Frontières** (Humanitarian Aid)

Sports: **Marathon** (2h46 PB, Bruges 2024) | **Rock Climbing** (Since age 4) | **Rugby** (9 yrs – Int. matches in Asia) | **Athletics** (800m → Marathon – 12-15h training / week)

Travel & Culture: **Expatriation Explorer** (3+ yrs in Bangkok, Thailand) | **Nature Excursions / Hiking**

Other: **Literature and Philosophy** | **Cultural Immersion** (High interest in foreign cultures)